

WEEK 5 PRACTICAL

QUESTION 1

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Aim

To implement communication between a parent and child process using an anonymous pipe and measure the communication time.

Program

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

#define COUNT 100000

int main() {
    int fd[2];

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        // Child - Consumer
        close(fd[1]);

        int data;
        int received = 0;

        while (read(fd[0], &data, sizeof(data)) > 0) {
            received++;
        }

        close(fd[0]);
    }
```

```

        printf("Consumer received: %d values\n", received);
    }
    else {
        // Parent - Producer
        close(fd[0]);

        int data;
        struct timespec start, end;

        clock_gettime(CLOCK_MONOTONIC, &start);

        for (int i = 0; i < COUNT; i++) {
            data = i;
            write(fd[1], &data, sizeof(data));
        }

        close(fd[1]);

        clock_gettime(CLOCK_MONOTONIC, &end);

        double time_taken =
            (end.tv_sec - start.tv_sec) +
            (end.tv_nsec - start.tv_nsec) / 1e9;

        printf("Producer generated: %d values\n", COUNT);
        printf("Communication time: %.6f seconds\n", time_taken);

        wait(NULL);
    }

    return 0;
}

```

Output

```

● hrushika@Hrushikas-MacBook-Air ossp practise % ./week_5
Consumer (PID 4767): received 1000000 integers, data verified.

Producer (PID 4764): sent 1000000 integers in blocks of 1024.
Total data transferred : 3.81 MB
Total communication time: 0.004097 seconds
Throughput             : 931.10 MB/s
❖ hrushika@Hrushikas-MacBook-Air ossp practise % █

```

Observation

The parent process generated data and sent it through the pipe. The child process successfully received the data, and the communication time was measured.

Result

The producer-consumer communication system was successfully implemented using an anonymous pipe.

QUESTION 2

Develop a program that executes the equivalent of the shell command `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Aim

To implement a pipeline between two commands using `fork()`, `pipe()`, `dup2()` and `exec()`.

Program

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    pid_t pid1 = fork();

    if (pid1 == 0) {
        // First child: ls -l
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("exec ls failed");
        exit(1);
    }

    pid_t pid2 = fork();
```

```

    if (pid2 == 0) {
        // Second child: grep ".c"
        close(fd[1]);

        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("grep", "grep", ".c", NULL);

        perror("exec grep failed");
        exit(1);
    }

    // Parent
    close(fd[0]);
    close(fd[1]);

    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);

    printf("Command pipeline completed.\n");

    return 0;
}

```

Output

```

hrushika@hrushikas-MacBook-Air ossp practise % gcc week52.c -o week52
hrushika@Hrushikas-MacBook-Air ossp practise % ./week52
-rw-r--r--@ 1 hrushika  staff   2099 Sep 19 13:35 week52.c
-rw-r--r--@ 1 hrushika  staff   4827 Sep 19 13:32 week_5.c
❖ hrushika@Hrushikas-MacBook-Air ossp practise %

```

Observation

The output of `ls -l` was passed through the pipe to `grep ".c"`, which displayed the matching C files.

Result

The shell command `ls -l | grep ".c"` was successfully implemented using system calls.